

A cube with a side length of 5 cm floats on the surface of water. If 4/5 of the cube volume floats above the surface of the water, what is the density of the cube? (Note: Density of water = 1,000 kg/m³.)

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A. 200 kg/m³ [51%]

B. 400 kg/m³ [8%]

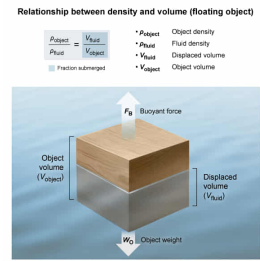
C. 800 kg/m³ [36%]

D. 2,500 kg/m³ [3%]

Correct

52% answered correctly

Explanation:



Archimedes' principle states that the magnitude of the buoyant force F_B that acts on a floating object is equal to the weight of the fluid displaced by the object. Consequently, F_B equals the product of the acceleration of gravity g and the fluid's density ρ_{fluid} and displaced volume V_{fluid} :

$$F_B = \rho_{\text{fluid}} \cdot V_{\text{fluid}} \cdot g$$

Furthermore, F_B has an equal magnitude and opposite direction to the floating object's weight W_O :

$$F_B = W_O$$

W_O equals the product of the mass m_{object} and g :

$$W_O = m_{\text{object}} \cdot g$$

Because m_{object} is the product of the object's density ρ_{object} and volume V_{object} , W_O is equal to:

$$W_O = \rho_{\text{object}} \cdot V_{\text{object}} \cdot g$$

Combining the equations for F_B and W_O gives:

$$\begin{aligned} \rho_{\text{fluid}} \cdot V_{\text{fluid}} \cdot g &= \rho_{\text{object}} \cdot V_{\text{object}} \cdot g \\ \rho_{\text{fluid}} \cdot V_{\text{fluid}} &= \rho_{\text{object}} \cdot V_{\text{object}} \end{aligned}$$

Rearranging the equation above yields a relationship between the ratios of the densities and volumes:

$$\frac{\rho_{\text{object}}}{\rho_{\text{fluid}}} = \frac{V_{\text{fluid}}}{V_{\text{object}}}$$

The fraction of the object's volume x that is submerged in the fluid equals the ratio of V_{fluid} and V_{object} , which also equals the ratio of ρ_{object} and ρ_{fluid} , as derived above:

$$x = \frac{V_{\text{fluid}}}{V_{\text{object}}} = \frac{\rho_{\text{object}}}{\rho_{\text{fluid}}}$$

In this question, 4/5 of V_{object} floats above the surface of the water. Therefore, 1/5 of V_{object} is submerged below the water surface, and the cube displaces a volume of water that is 1/5 its own volume:

$$\frac{1}{5} = \frac{V_{\text{fluid}}}{V_{\text{object}}} = \frac{\rho_{\text{object}}}{\rho_{\text{fluid}}}$$

Given that ρ_{fluid} is 1,000 kg/m³, ρ_{object} equals:

$$\begin{aligned} \frac{\rho_{\text{object}}}{1,000 \text{ kg/m}^3} &= \frac{1}{5} \\ \rho_{\text{object}} &= \frac{1}{5} \cdot (1,000 \text{ kg/m}^3) = 200 \text{ kg/m}^3 \end{aligned}$$

Therefore, the density of the cube is 200 kg/m³.

(Choice B) A cube with a density of 400 kg/m³ would float with 3/5 (not 4/5) of the cube volume above the water.

(Choice C) A cube with a density of 800 kg/m³ would float with 1/5 (not 4/5) of the cube volume above the water.

(Choice D) A cube with a density of 2,500 kg/m³ is denser than water and would not float at all.

Educational objective:

Archimedes' principle states that the buoyant force acting on a floating object is equal to the weight of the fluid displaced by the object. The fraction of the object submerged is related to the density of the object and the density of the fluid in which the object floats.

Time spent:0 sec
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Physics

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